

AI INVESTMENT SIMULATOR

Purple City Investment Foundations

A visual teaching deck for Portals 1–9

\$ STARTING CAPITAL

\$14,800 virtual cash

➤ LEARNING MODE

Learn → Research → Decide

✓ GOAL

Defend a disciplined first portfolio

Your Purple City journey

Nine portals. One evolving portfolio. Each portal adds a skill.



The portfolio does not reset. Every concept changes how you evaluate the same \$14,800.

PORTAL 1

Enter the Market

Understand saving, investing, trading, risk, return, compounding, inflation and real returns.



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Saving, investing and trading

They all use money differently — and solve different problems.

▣ SAVING

Purpose: preserve capital and keep it accessible.

Typical horizon: short.

Main concern: inflation may reduce purchasing power.

➤ INVESTING

Purpose: build wealth or income over time.

Typical horizon: medium to long.

Main concern: market and business risk.

↔ TRADING

Purpose: profit from shorter-term price moves.

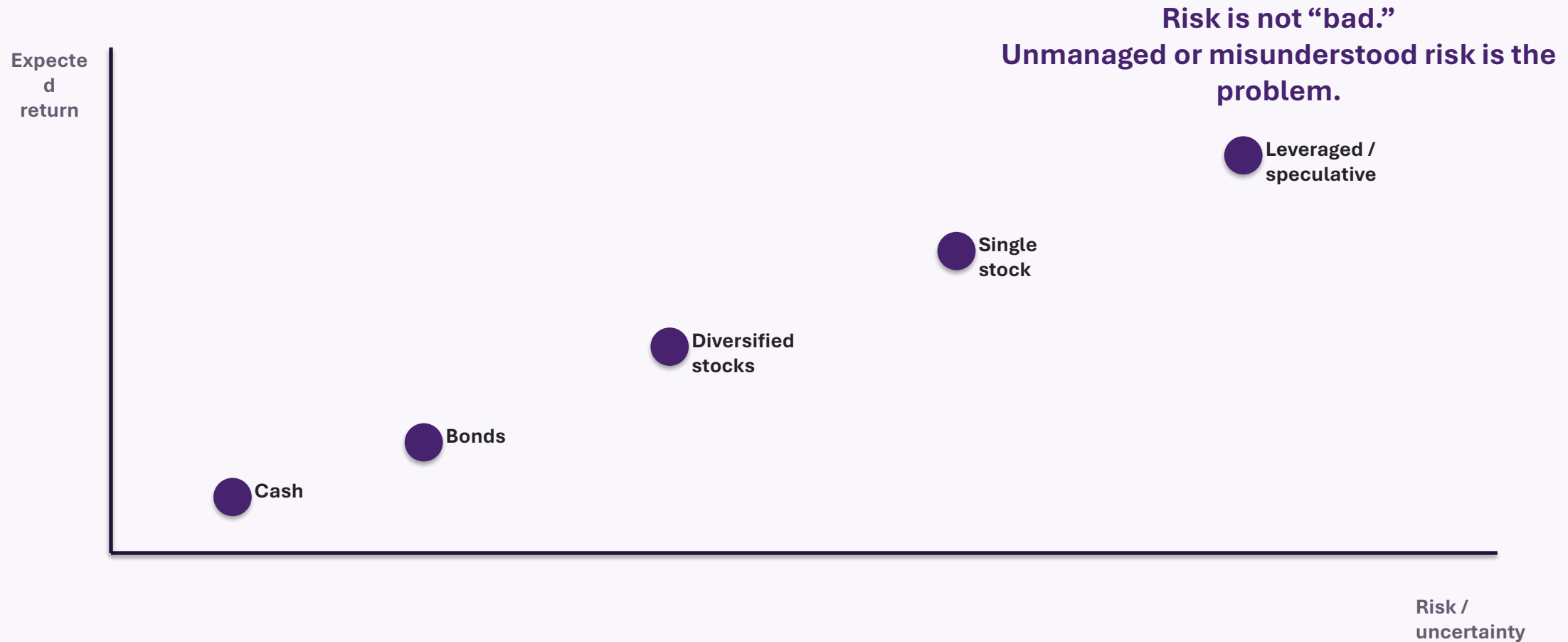
Typical horizon: short.

Main concern: timing, volatility and transaction costs.

Beginner rule: first decide what the money is for. The purpose should drive the strategy.

Risk and return: the central trade-off

Higher expected return normally requires accepting more uncertainty.



Present value and future value

Money has a time dimension: today's dollars can earn returns.

Future value

$$FV = PV(1 + r)^n$$

Present value

$$PV = FV / (1 + r)^n$$

P **PV — Present Value**

What a future amount is worth today.

% **r — Rate**

Return or interest rate per compounding period.

n **n — Periods**

How many times the return compounds.

FV moves money forward in time. PV discounts money backward in time.

Compounding: returns begin earning returns

Example: \$2,000 invested at 8% annually for three years.



Worked PV/FV examples

Use the same formulas in different directions.

1 • ONE YEAR

\$1,000 at 5%
 $FV = \$1,000(1.05)$
 $= \$1,050$

2 • PRESENT VALUE

Need \$5,000 in 4 years at 6%
 $PV = \$5,000/(1.06)^4$
 $= \$3,960.47$

3 • MONTHLY COMPOUNDING

\$3,000 at 6%, monthly, 2 years
 $FV = \$3,000(1+.06/12)^{24}$
 $= \$3,381.48$

4 • YOUR \$14,800

At 7% for 5 years:
 $FV = \$14,800(1.07)^5$
 $= \$20,757.57$

KEY IDEA

Compounding frequency changes how often returns are added to the investment base.

Nominal return vs real return

A bigger account balance does not always mean greater purchasing power.

\$ NOMINAL RETURN

The percentage gain or loss measured in current money — before inflation.

Nominal return

$$\frac{(\text{Ending} - \text{Beginning} + \text{Income})}{\text{Beginning}}$$

🎯 REAL RETURN

The return after adjusting for inflation — the change in purchasing power.

Real return

$$(1 + \text{nominal}) / (1 + \text{inflation}) - 1$$

Nominal asks: “How many more dollars?” • Real asks: “What can those dollars actually buy?”

Five real-return examples

Inflation can change the meaning of an investment gain.

	Nominal 8%	Inflation 3%	Real 4.85%	$\frac{(1+0.08)}{(1+0.03)} - 1$
	Nominal 4%	Inflation 6%	Real -1.89%	$\frac{(1+0.04)}{(1+0.06)} - 1$
	Nominal 10%	Inflation 8%	Real 1.85%	$\frac{(1+0.10)}{(1+0.08)} - 1$
	Nominal 5%	Inflation 2%	Real 2.94%	$\frac{(1+0.05)}{(1+0.02)} - 1$
	Nominal 8%	Inflation 4%	Real 3.85%	$\frac{(1+0.08)}{(1+0.04)} - 1$

Example 2 is the warning: +4% nominal return still loses purchasing power when inflation is 6%.

Know Your Investor

Build an investment mandate around risk tolerance, risk capacity, time horizon and liquidity.



Risk tolerance is not risk capacity

One is psychological. The other is financial.

♥ RISK TOLERANCE

How much uncertainty and loss can you emotionally tolerate?

Example: Could you watch a portfolio fall 20% without abandoning the plan?

\$ RISK CAPACITY

How much loss can your finances actually absorb?

Example: Money needed for tuition next year has low capacity for market loss.

A person can be emotionally fearless but financially unable to take large risk — or cautious despite having high financial capacity.

Build your investment mandate

The mandate turns preferences into rules for the \$14,800.

Objective

What is the money trying to achieve?

Horizon

When might the money be needed?

Liquidity

How much must remain accessible?

Risk limits

How much loss/concentration is acceptable?

Asset rules

What may or may not be owned?

Review

When will decisions be reassessed?

Checkpoint vocabulary: “sell” = dispose of an investment you own in exchange for cash.

Understanding Asset Classes

Learn what you actually own, what can make it earn, and what can make it lose.



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Asset classes behave differently

Diversification starts by understanding different economic exposures.

Stocks

Company ownership

Moves with: Profits • growth • expectations

Bonds

Debt claims

Moves with: Rates • inflation • credit

ETFs

Baskets of assets

Moves with: Depends on holdings

Real Estate

Property exposure

Moves with: Rent • rates • location

Forex

Currency pairs

Moves with: Rates • policy • macro

Commodities

Physical resources

Moves with: Supply • demand • shocks

Cash

Liquid funds

Moves with: Rates • inflation

Businesses

Whole company

Moves with: Cash flow • execution

What tends to move bonds?

Bond prices and yields react to interest rates, inflation and credit risk.

MARKET RATES ↑ ————— **EXISTING BOND PRICE ↓**

INTEREST RATES

New bonds may offer higher yields, making older lower-coupon bonds less attractive.

INFLATION

Higher expected inflation reduces the purchasing power of fixed payments.

CREDIT QUALITY

If a borrower appears less able to repay, investors demand more yield and the bond price may fall.

Also watch maturity: longer-duration bonds are generally more sensitive to interest-rate changes.

ETFs: one trade, many holdings

An exchange-traded fund is a pooled investment whose shares trade on an exchange.



WHAT AN ETF CAN HOLD

- Hundreds of stocks
- Government or corporate bonds
- A sector such as technology
- Commodities or other exposures

Many ETFs track an index.



WHAT TO CHECK

Holdings
Objective/index
Expense ratio
Liquidity
Tracking difference
Concentration
Currency exposure



WHY INVESTORS USE THEM

Diversification
Convenience
Transparent exposure
Potentially low costs
Easy exchange trading

How real estate investing works

Returns can come from income, price appreciation — or both.

\$ INCOME ENGINE

Rent
– vacancy
– operating expenses
= property operating income

🏠 VALUE DRIVERS

Location • local supply
Employment • population
Interest/mortgage rates
Comparable property sales

! RISKS

Vacancy • maintenance
Leverage • refinancing
Illiquidity • taxes
Local economic weakness

Direct property vs REITs

DIRECT PROPERTY

Physical ownership; greater control, but slower and costlier to buy/sell.

REIT

Exchange-traded property exposure; more liquid, but market-priced like a security.

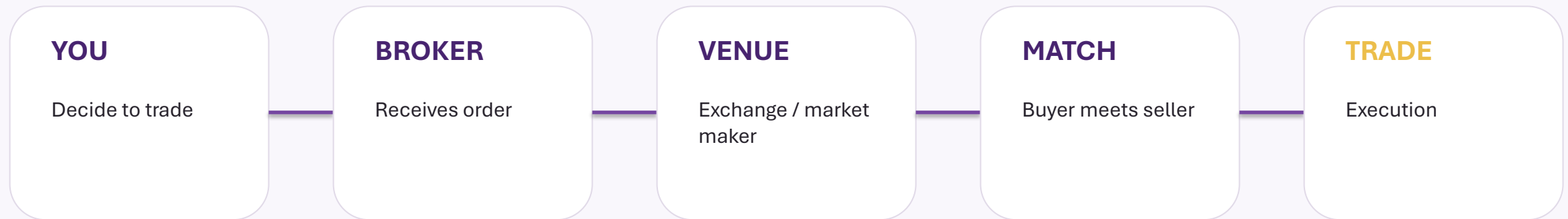
How Markets Work

Understand the route from investor → broker → market, then learn bid/ask, liquidity and order types.



From your decision to an executed trade

Markets are systems for matching buyers and sellers.



Bid = highest current buying offer. Ask = lowest current selling offer. Spread = Ask – Bid.

$$\text{\$50.05} - \text{\$49.95} = \text{\$0.10 spread}$$

What creates liquidity?

Liquidity is the ability to trade quickly without a large price concession.

✓ MORE LIQUID

Narrower spreads
Deeper order book
Less price impact

Usually supported by:

- Many willing buyers and sellers
- Frequent trading
- Orders close to current price
- Accessible information
- Market-making activity

! LESS LIQUID

Wider spreads
More slippage
Greater price impact

Can occur with:

- Few participants
- One-sided panic or euphoria
- Obscure securities
- Large order relative to market
- Sudden uncertainty

Liquidity is not fixed. It can disappear quickly when market conditions change.

Market order vs limit order

Execution certainty and price control pull in different directions.

▶ MARKET ORDER

Instruction: trade now at the best available prices.

Priority: execution.

Risk: final price may differ from what you saw.

|| LIMIT ORDER

Instruction: trade only at your limit price or better.

Priority: price control.

Risk: the order may never execute.

Example order book: ABC bid \$49.90 • ask \$50.10

BUY MARKET

Begins near \$50.10

BUY LIMIT \$49.50

Waits for price $\leq$ \$49.50

BUY LIMIT \$50.10

May execute immediately

Trading has visible and invisible costs

Small frictions matter when repeated.

SPREAD

Buy 100 at \$20.10, sell at \$20.00

$$100 \times \$0.10 = \$10$$

COMMISSION

\$5 to buy + \$5 to sell

\$10 total

SLIPPAGE

Expected \$30.00; filled \$30.08 on 200 shares

$$200 \times \$0.08 = \$16$$

MARKET IMPACT

Large order consumes cheaper offers

Average price rises

Frequent activity can erode returns even when each individual cost looks small.

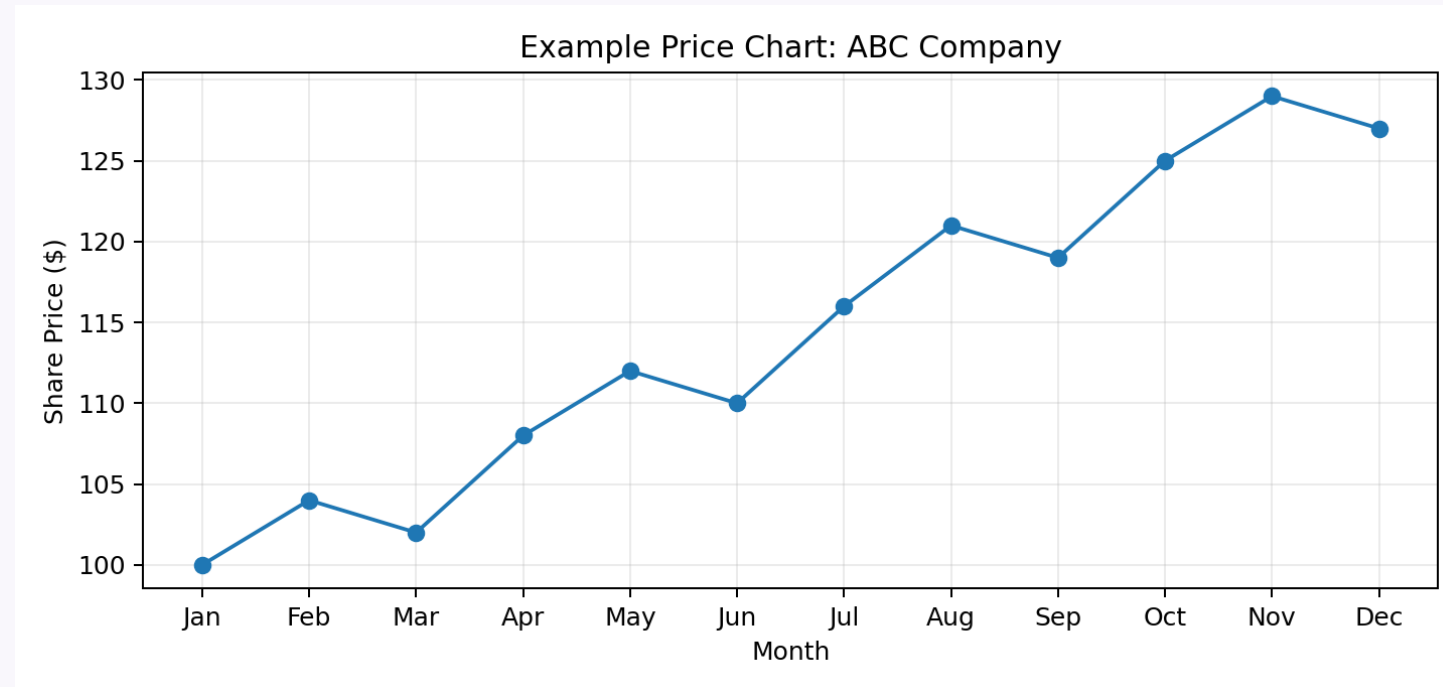
Reading the Market

Interpret charts, market capitalization, indices, volume, volatility and sentiment — without confusing price with value.



Read the chart before explaining the chart

First describe what happened. Then investigate why.



? BEGINNER READING

1. Direction: up, down or sideways?
2. Range: high and low?
3. Pullbacks: where did price retreat?
4. Variability: smooth or unstable?
5. Then ask: what information changed?

A price chart is evidence about historical behavior — not proof of fundamental value or future direction.

Market capitalization: how large is the equity value?

A share price alone does not tell you how large a company is.

$$\text{Market capitalization} = \text{Share price} \times \text{Shares outstanding}$$

EXAMPLE 1

10 million shares $\times$ \$25
= \$250 million

EXAMPLE 2

500 million shares $\times$ \$80
= \$40 billion

KEY COMPARISON

\$200 $\times$ 1m shares = \$200m
\$20 $\times$ 50m shares = \$1bn

Higher share price $\neq$ larger company. Market cap measures equity value; it is not the same as the value of the whole enterprise including debt and cash.

Indices: the market's measuring sticks

An index represents a defined group of securities and can serve as a benchmark.

S&P 500

Large U.S. companies

DJIA

30 prominent U.S. companies

Nasdaq Composite

Nasdaq-listed; tech-heavy

FTSE 100

Large UK-listed companies

MSCI World

Developed-market global equities

Use indices to compare performance, describe market direction and select a relevant benchmark.

MARKET-CAP WEIGHTED

Larger companies have more influence.

PRICE WEIGHTED

Higher-priced shares have more influence.

Volume, volatility and sentiment

Three different lenses on market behavior.

VOLUME

How much changes hands.

High volume can signal strong participation in a move.

But volume alone does not tell you whether the asset is good value.

VOLATILITY

How widely returns or prices fluctuate.

Higher volatility = historically larger swings.

Useful risk indicator, but not identical to permanent-loss risk.

SENTIMENT

Market optimism, pessimism or risk appetite.

Can be inferred from surveys, flows, options, news and price behavior.

Sentiment can move price without changing fundamentals.

Sample volatility of returns

$$s = \sqrt{[\Sigma(r_i - \bar{r})^2 / (n - 1)]}$$

Market reaction is not the same as fundamental information

Prices respond to information relative to expectations.

ABC rises 8% in one day • volume is 3× normal • earnings beat expectations • management raises guidance

FUNDAMENTALS

Earnings and guidance changed.

MARKET REACTION

Price rose sharply and trading activity surged.

SENTIMENT

Optimism may have increased.

INVESTOR'S NEXT JOB

Check valuation, sustainability and what was already expected.

A company can report good results and still fall if investors expected even better results.

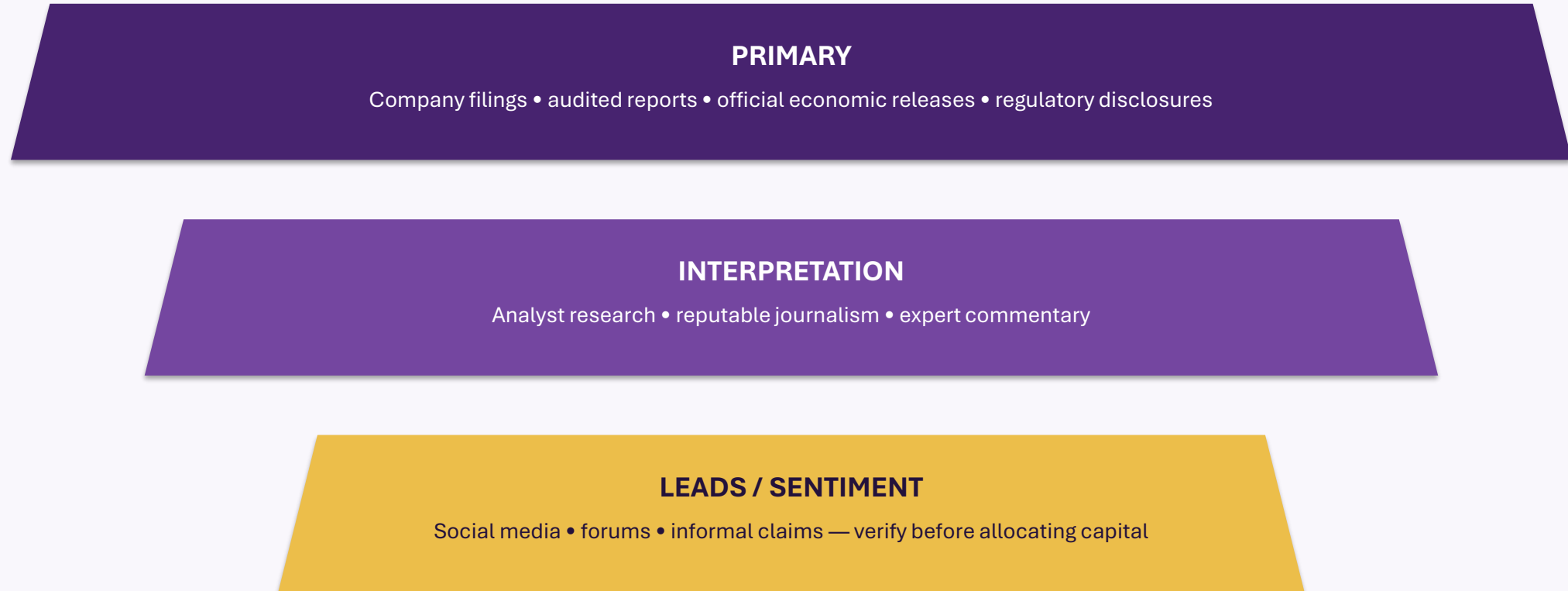
Research Fundamentals

Learn to separate evidence from commentary and build a research process that can challenge your own beliefs.



Build an evidence hierarchy

Start close to the underlying facts, then add interpretation.



Research question: “What would I need to know to decide whether this deserves capital?”

Six questions for source credibility

A source can be useful without being equally reliable for every claim.

1 Authority

Who produced it?

2 Evidence

Is the claim supported?

3 Timeliness

Is it current enough?

4 Incentives

Who benefits if you believe it?

5 Consistency

Do credible sources agree?

6 Specificity

Can the claim be tested?

Good research actively searches for evidence that could prove the investment idea wrong.

Build Your First Portfolio

Turn research into position sizes, portfolio weights, diversification and a deliberate cash allocation.



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Position size is a risk decision

Even a good idea can damage a portfolio if the exposure is too large.

$$\text{Portfolio weight} = \text{Position value} / \text{Total portfolio value} \times 100\%$$

Stocks

\$5,180 • 35%

Bonds

\$2,960 • 20%

ETF

\$2,220 • 15%

Real estate

\$1,480 • 10%

Commodity

\$740 • 5%

Cash

\$2,220 • 15%

Illustrative allocation only — the learning goal is to defend the logic, not copy a “correct” portfolio.

Diversification is about different risks — not just more holdings

Ten investments can still be one concentrated bet.

🎯 LOOK BEYOND NUMBER OF HOLDINGS

Check concentration by:

- Asset class
- Sector / industry
- Geography
- Currency
- Company-specific exposure
- Liquidity

\$ WHY HOLD CASH?

Cash can:

- Meet near-term needs
- Reduce short-term volatility
- Preserve optionality
- Fund future opportunities

Cost: inflation and foregone return.

Diversification does not eliminate loss. It aims to avoid relying too heavily on one source of risk.

Investment Decisions

Practice BUY, ADD, HOLD, TRIM and EXIT — and document why capital should stay or move.



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Five active portfolio decisions

“Do nothing” can still be a deliberate investment decision.

BUY

Open a new position

Typical reason:
Reward justifies risk

ADD

Increase exposure

Typical reason:
Evidence strengthens
/ valuation improves

HOLD

Keep unchanged

Typical reason:
Thesis intact; size
appropriate

TRIM

Reduce exposure

Typical reason:
Risk, concentration
or valuation rises

EXIT

Close position

Typical reason:
Thesis breaks / better
use of capital

Opportunity cost: every dollar committed here is a dollar unavailable for the best alternative opportunity.

The decision journal turns action into learning

A good rationale can be reviewed later — even when the outcome is bad.

ACTION

BUY / ADD / HOLD / TRIM / EXIT

EVIDENCE

What changed? What supports the action?

EXPECTED REWARD

What outcome are you being paid to pursue?

MAIN RISK

What could cause loss?

NEW WEIGHT

How large is the exposure now?

FALSIFIER

What would prove the decision wrong?

“The price is going up” is an observation — not a complete investment thesis.

Purple City Challenge

Defend the process behind the portfolio, identify weaknesses, and rebalance before Blue City.



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How to defend your first portfolio

The challenge rewards disciplined reasoning — not lucky short-term returns.

1 • Mandate

Objective • horizon • risk • liquidity

2 • Allocation

Asset weights • cash reserve

3 • Holdings

Purpose • thesis • size

4 • Evidence

Strongest sources • uncertainty

5 • Risk

Concentrations • downside scenarios

6 • Decision quality

Explain an active decision

7 • Weaknesses

What is missing or poorly supported?

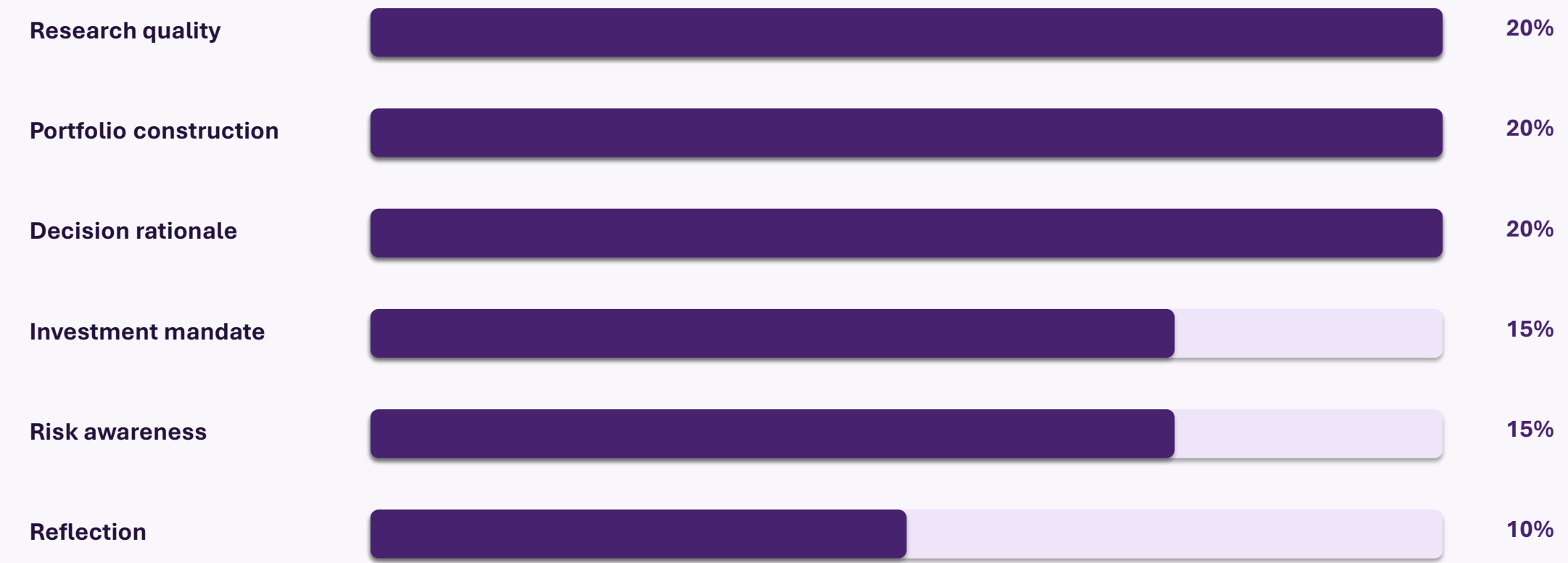
8 • Rebalance

What must change before Blue City?

A defensible portfolio can still lose money. The question is whether the decisions were evidence-based and risk-aware.

Purple City scoring rubric

What strong work should demonstrate.



Highest weight: research, construction and decision rationale. Performance alone is not the grade.

Before you unlock Blue City...

- ✓ Explain saving vs investing vs trading
- ✓ Calculate PV, FV, nominal and real returns
- ✓ Describe major asset classes and what moves them
- ✓ Explain bid/ask, liquidity, market and limit orders
- ✓ Read basic market data without treating charts as certainty
- ✓ Evaluate sources and maintain a research log
- ✓ Calculate and defend portfolio weights
- ✓ Use BUY, ADD, HOLD, TRIM and EXIT deliberately
- ✓ Defend and rebalance the \$14,800 portfolio

Next: BLUE CITY — Analysis & Asset Selection • “What should I invest in, and what is it worth?”